Lesson plan · for the teacher

Decisions: if, elif and else

Programming · Level 2 · Lesson 2

Time
40 min

Materials
Board and chalk; Cards with made-up marks, including 49, 50, 69 and 70; Your phone with this lesson downloaded

Learning goals

- Compare values with ==, <, >= and the others.
- Write if, elif and else with colons and indents.
- Test a marking rule at its boundaries.

Activity: Find your corner

1. Warm-up (5 min): Call out conditions: “Stand up if you wear black shoes”, “...if your birthday is in May or June”. Agree that a condition is either true or false.
2. Teach (10 min): Write the grading rule on the board as if, elif and else. Show the colon and the indent. Ask: why must “70 or more” come before “50 or more”?
3. Sort (10 min): Give each pupil a mark card, including 49, 50, 69 and 70. Label three corners A, B and C. Pupils go to their corner; the class checks each one with the rule.
4. Run (10 min): Type the marks that caused arguments into the program on your phone and read out the grades. Did the program agree with the class?
5. Wrap-up (5 min): Ask which marks were hardest, and why boundaries cause bugs. Then give the exit ticket.

One phone is enough: you run the program, and the pupils are the data. Download the lesson the day before so Python works offline.

Discussion

- Why is 50 the mark most likely to be graded wrong?
- What goes wrong if the B test comes before the A test?
- Which other school rule could be an if statement?

Exit ticket

- Write the marks you would test for “40 or more passes”, and why.

Try it in class tomorrow
Write your school’s grading rule on the board as if, elif and else. Give pupils five marks, including 49 and 50, and ask which grade each one gets. Then run them on your phone: did the program agree?

Worksheet · for pupils

Decisions: if, elif and else

Programming · Level 2 · Lesson 2

Name _____ Class _____ Date _____

1 **A pupil writes this line with one equals sign. What happens when the program runs?**

```
if mark = 50:
    print("Pass")
```

◯ It works the same as == ◯ Python stops with a SyntaxError

◯ It sets mark to 50, then prints Pass

2 **Finish the grading program: 70 or more prints A, 50 or more prints B, anything else prints C. It is tested at the boundaries.**

```
mark = int(input("Mark? "))
# print A for 70 or more, B for 50 or more, else C
```

Write your program:

3 **In an if … elif … else, Python runs every branch whose condition is true.**

◯ True ◯ False

4 **Put the lines in order: the program asks for a mark and prints Pass for 50 or more, and Try again for anything less. One line doesn't belong.**

Number the lines in order (1, 2, 3…). Cross out any line that doesn't belong.

☐ `if mark >= 50:`

☐ `if mark > 50:`

☐ `mark = int(input("Mark? "))`

☐ `print("Try again")`

☐ `else:`

☐ `print("Pass")`

5 **The rule is “70 or more is A, 50 or more is B, else C”, but a pupil with exactly 50 gets C. Which line is wrong?**

Circle the line with the mistake. Write it correctly:

```
mark = int(input("Mark? "))
if mark >= 70:
    print("A")
elif mark > 50:
    print("B")
else:
    print("C")
```

✎ **Which rule in your school could a program apply for you: grades, fees, attendance? Write it as if, elif and else, and list the boundary values you would test.**
